

Problem Sheet 2

Exercise 1 (Measure Theory and Probability Densities)

Solution: The mistake is that the additivity property only works for countably many sets, not for uncountably many sets. In a continuous distribution, each single point has probability 0, but the interval contains uncountably many points. Therefore, we cannot sum the probabilities of all the single points in the same way. The probability measure is only countably additive.

Exercise 2 (Moments)

Proof: Since $X \sim \mathcal{N}(\mu, \sigma^2)$, its density is

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

We use the substitution

$$y = \frac{x-\mu}{\sigma}, \quad x = \mu + \sigma y, \quad dx = \sigma dy.$$

First, we compute the expectation:

$$\mathbb{E}[X] = \int_{\mathbb{R}} x f_X(x) dx.$$

Since $x = \mu + \sigma y$, we get

$$\mathbb{E}[X] = \int_{\mathbb{R}} (\mu + \sigma y) \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy.$$

Thus

$$\mathbb{E}[X] = \mu \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy + \sigma \int_{\mathbb{R}} y \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy.$$

The first integral is 1, and the second integral is 0.

Therefore,

$$\mathbb{E}[X] = \mu.$$

we compute the variance:

$$\mathbb{V}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

Since $\mathbb{E}[X] = \mu$, we have

$$\mathbb{V}[X] = \mathbb{E}[(X - \mu)^2] .$$

Hence

$$\mathbb{V}[X] = \int_{\mathbb{R}} (x - \mu)^2 f_X(x) dx .$$

Since,

$$x - \mu = \sigma y .$$

Therefore,

$$\mathbb{V}[X] = \int_{\mathbb{R}} \sigma^2 y^2 \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy .$$

So

$$\mathbb{V}[X] = \sigma^2 \int_{\mathbb{R}} y^2 \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy .$$

By the given hint,

$$\int_{\mathbb{R}} y^2 \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy = 1 .$$

Thus

$$\mathbb{V}[X] = \sigma^2 .$$

Therefore,

$$\mathbb{E}[X] = \mu \quad \text{and} \quad \mathbb{V}[X] = \sigma^2 .$$

sht2_ex3

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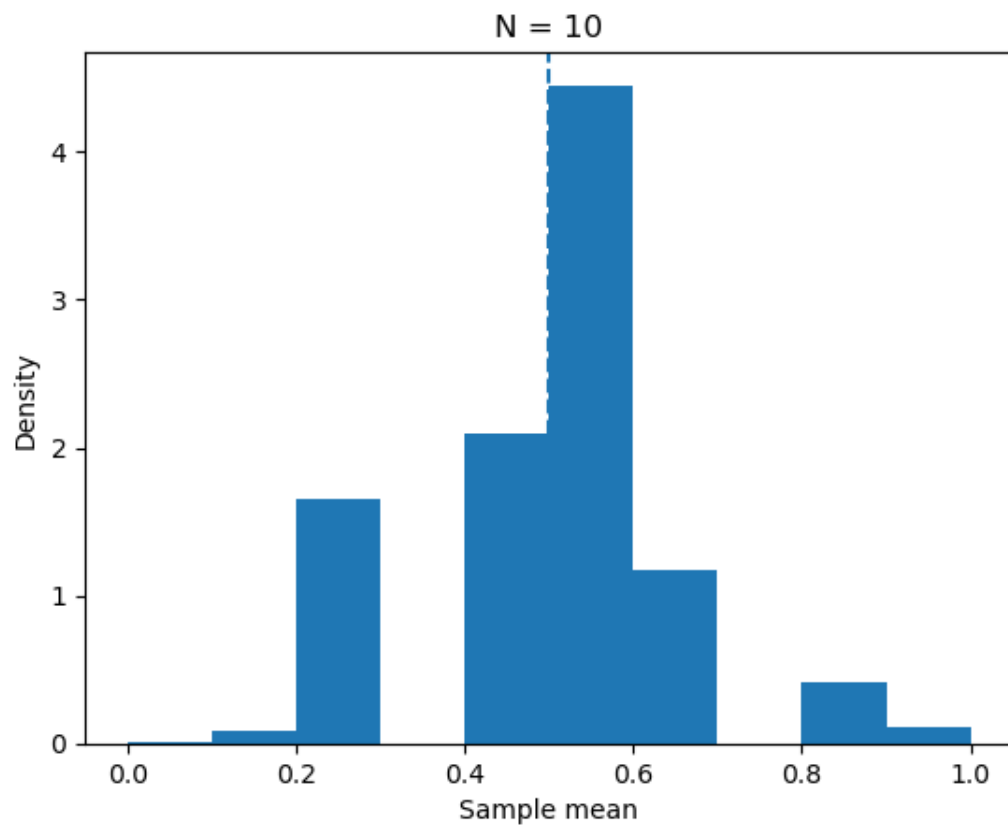
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[1]: import numpy as np
import matplotlib.pyplot as plt

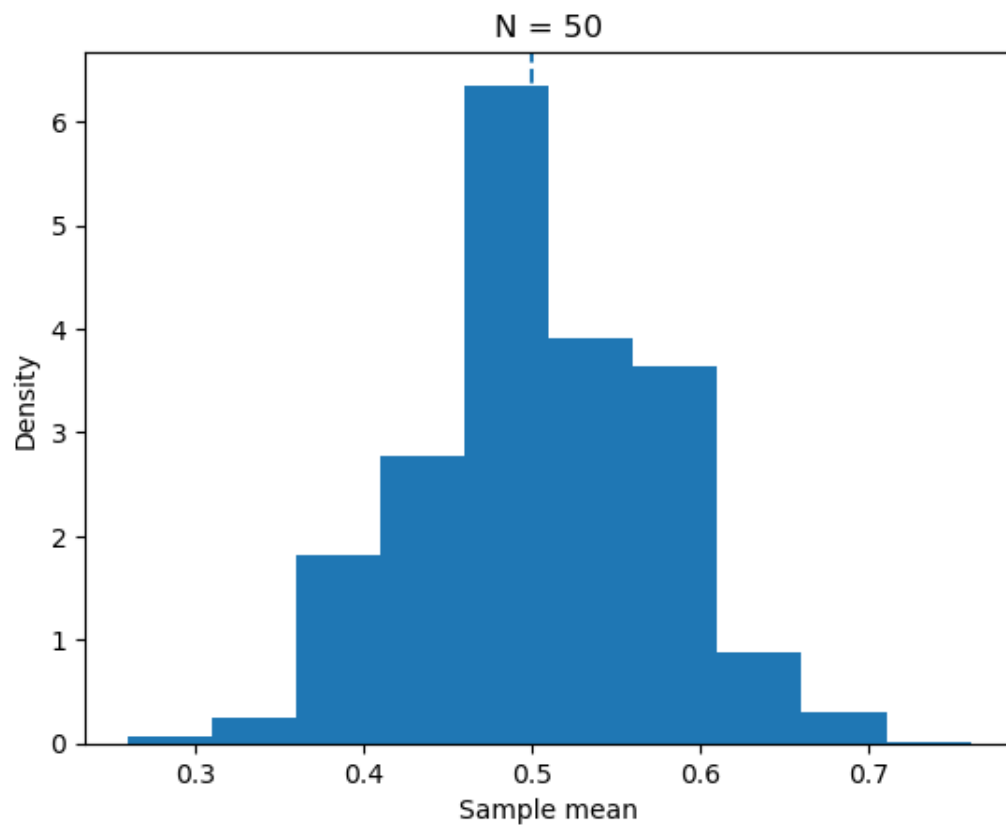
[4]: sample_sizes = [10, 50, 100]
p = 0.5
epoch = 10000

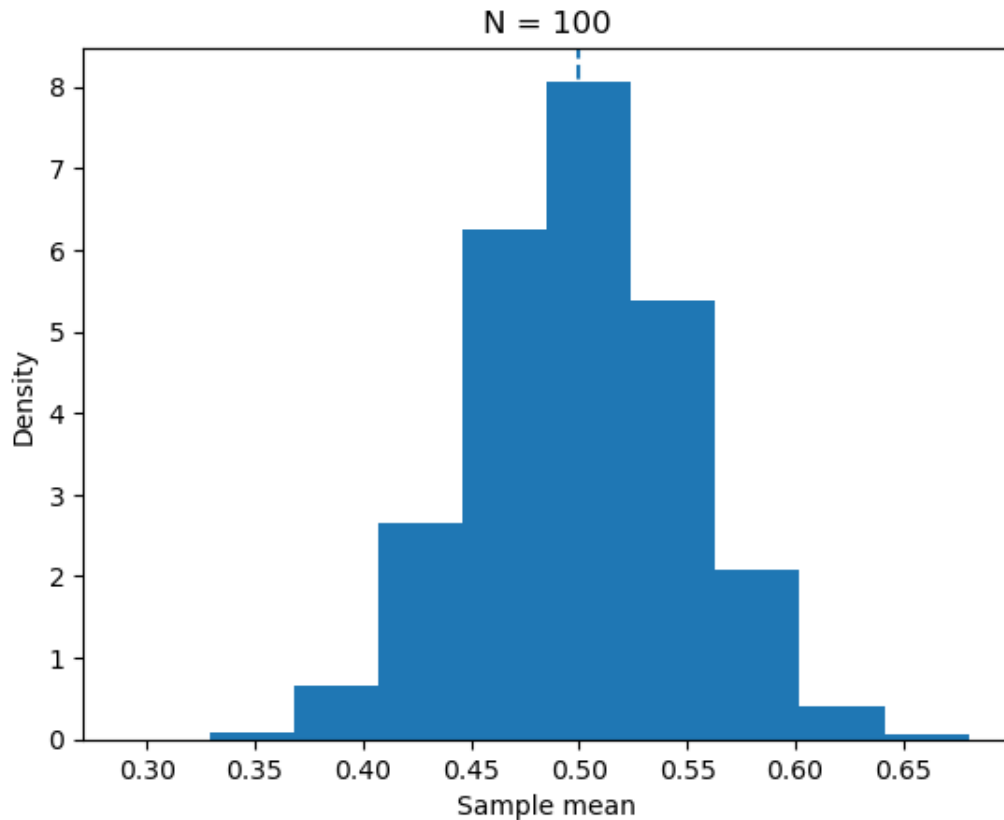
[7]: for N in sample_sizes:
    means = []
    for r in range(epoch):
        sample = np.random.binomial(1, p, N)
        sample_mean = np.mean(sample)
        means.append(sample_mean)

    # Plot histogram
    plt.hist(means, density=True)
    plt.title("N = " + str(N))
    plt.xlabel("Sample mean")
    plt.ylabel("Density")
    plt.axvline(0.5, linestyle="--")

    plt.show()
```







0.1 Observations:

We observe that the histograms are centered around 0.5. For $N = 10$, the histogram is wider. For $N = 50$ and $N = 100$, the histogram becomes narrower and more bell-shaped. This shows as the sample size increases, the distribution of the sample mean approaches a normal distribution and becomes more concentrated around the true mean which illustrate the Central limit theorem.

[]:

Ex 4

1.

$$\log(\mu_i) = \beta_0 + \beta_1 \cdot x_i$$

$$\mu_i = e^{\beta_0 + \beta_1 x_i}$$

$$\begin{aligned} L(\beta_0, \beta_1) &= \prod_{i=1}^N \frac{\mu_i^{c_i}}{c_i!} e^{-\mu_i} \\ &= \prod_{i=1}^N \frac{(e^{\beta_0 + \beta_1 x_i})^{c_i}}{c_i!} e^{-e^{\beta_0 + \beta_1 x_i}} \end{aligned}$$

$$\log(\beta_0, \beta_1) = \sum_{i=1}^N [c_i (\beta_0 + \beta_1 x_i) - e^{\beta_0 + \beta_1 x_i} - \log(c_i!)]$$

2.

$$\frac{\partial \log}{\partial \beta_0} = \sum_{i=1}^N (c_i - e^{\beta_0 + \beta_1 x_i}) = 0.$$

$$\begin{aligned} \sum_{i=1}^N c_i &= \sum_{i=1}^N e^{\beta_0 + \beta_1 x_i} \\ &= \sum_{i=1}^N e^{\beta_0} e^{\beta_1 x_i} \\ &= e^{\beta_0} \sum_{i=1}^N e^{\beta_1 x_i} \end{aligned}$$

$$\beta_0 = \log\left(\frac{\sum_{i=1}^N c_i}{\sum_{i=1}^N e^{\beta_1 x_i}}\right)$$